

A support-product minimizer whose supports are not difference sets

Disproof of Conjecture 00000009278

gaochengzhecpu

Abstract

For complex functions on $G = \mathbb{Z}/4\mathbb{Z}$, the smallest nonzero Fourier support product is 4. The indicator of $H = \{0, 2\}$ attains this minimum, and both its support and its Fourier support equal H . This set is not a difference set. Thus the asserted difference-set classification of extremals is false, although the usual uncertainty lower bound remains valid. We prove global minimality over all nonzero complex functions, not just optimality within a test family.

Conventions and the claim being refuted

Identify the dual of $G = \mathbb{Z}/4\mathbb{Z}$ with G by its standard characters. For $f : G \rightarrow \mathbb{C}$, use the unnormalized Fourier transform

$$\widehat{f}(k) = \sum_{x \in G} f(x) e^{-2\pi i x k / 4}, \quad \text{supp } f = \{x \in G : f(x) \neq 0\}.$$

Changing to the normalized transform does not change either support. The support product is $|\text{supp } f| |\text{supp } \widehat{f}|$, for $f \neq 0$. An extremal here means a function attaining its global minimum.

A subset $D \subseteq G$ is a difference set if some integer $\lambda \geq 0$ satisfies

$$\#\{(a, b) \in D^2 : a - b = g\} = \lambda \quad \text{for every } g \in G \setminus \{0\}.$$

This definition even allows degenerate parameter choices. We disprove the source's assertion that support-product extremals are difference sets: our minimizing function has neither a difference-set support nor a difference-set Fourier support. A cyclic finite abelian group already lies within the stated finite-group scope.

The actual minimum is four

The elementary inversion identity is

$$\widehat{\widehat{f}}(x) = 4f(-x).$$

Indeed, expand the two sums: the inner sum over k is 4 if the frequency is zero modulo 4, and 0 otherwise. In particular $f \neq 0$ implies $\widehat{f} \neq 0$.

Lemma 1. *Every nonzero function $f : G \rightarrow \mathbb{C}$ satisfies $|\text{supp } f| |\text{supp } \widehat{f}| \geq 4$.*

Proof. Both support sizes are positive integers. If each is at least 2, their product is at least 4. If $\text{supp } f = \{j\}$, then

$$\widehat{f}(k) = f(j) e^{-2\pi i j k / 4} \neq 0 \quad \text{for every } k \in G,$$

so the sizes are 1 and 4. If $\text{supp } \widehat{f}$ is a singleton, the same argument applied to $\widehat{f}$ shows that $\widehat{\widehat{f}}$ is nonzero everywhere. Inversion then shows that f is nonzero everywhere. The sizes are 4 and 1. These cases exhaust all possibilities. $\square$

An extremal with two nondifference-set supports

Take the proper nontrivial subgroup $H = \{0, 2\}$ and $f = 1_H$. Directly from the transform definition,

$$\widehat{f}(k) = 1 + e^{-\pi i k} = 1 + (-1)^k = \begin{cases} 2, & k \in \{0, 2\}, \\ 0, & k \in \{1, 3\}. \end{cases}$$

Thus $f = (1, 0, 1, 0)$, $\widehat{f} = (2, 0, 2, 0)$, and

$$\text{supp } f = \text{supp } \widehat{f} = H, \quad |\text{supp } f| |\text{supp } \widehat{f}| = 2 \cdot 2 = 4.$$

The lemma proves that this is an actual global minimum.

However, the differences of the four ordered pairs in H^2 are

$$0 - 0 = 0, \quad 0 - 2 = 2, \quad 2 - 0 = 2, \quad 2 - 2 = 0 \pmod{4}.$$

Hence nonzero difference 1 occurs 0 times and nonzero difference 2 occurs 2 times. No common λ exists.

Theorem 1. *There is a nonzero complex function on a finite group attaining the minimum Fourier support product such that neither support is a difference set. Consequently the extremal-classification assertion in Conjecture 00000009278 is false.*

Proof. The function 1_H constructed above has all the stated properties. $\square$

Formal correspondence and reproduction

The Lean project works with the actual group `ZMod 4`, complex values, and Mathlib’s standard `ZMod.dft` linear equivalence. It uses the library’s Fourier inversion theorem, rather than replacing the transform by a custom table. The support predicate is precisely nonzero function values. The theorem `support_product_bound` quantifies over every nonzero complex function on the group.

The project constructs H as both a finite set and an additive subgroup. The theorem `fourier_witness` evaluates the actual transform and proves $\widehat{1_H} = 2 1_H$. Character values are derived from the standard character’s homomorphism and injectivity properties. Both support identifications and the product 4 are proved. The difference-set predicate counts actual ordered pairs with the specified group difference; the counts 0 and 2 are kernel checked. The final theorem `counterexample` asserts nonzeroness, global minimality, and the failure of the difference-set property for both supports.

Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. In `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The cache command obtains the pinned official dependency artifacts if absent. The submitted project is rebuilt in a fresh directory. No auxiliary numerical program, extra axiom, proof placeholder, or external computational oracle is used. Run `tectonic main.tex` to regenerate the PDF. The author performed a separate adversarial self-review; no independent reviewer or subagent was used.

Source

The Last Math Competition, Conjecture 00000009278. The exact bilingual statement is preserved in `SOURCE.md`; its upstream revision, source hash, actual build logs, axiom dependencies and review records accompany this submission.